Gradient Attribution and Attention-Based Explainability in Regime-Switching Financial Markets

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1 Executive Summary

This study extends existing interpretability frameworks along three complementary dimensions, with the objective of improving both model robustness and economic interpretability in regime-dependent environments.

First, we introduce a **data expansion and preprocessing strategy** to increase the effective sample size. By restructuring and augmenting the original data, this approach mitigates small-sample limitations that commonly arise in regime-specific modeling, leading to more stable estimation and learning without altering the underlying data-generating process.

Second, we incorporate **regime segmentation via R2RD**, enabling the model to explicitly account for structural heterogeneity across different market or economic regimes. Rather than assuming a single stationary relationship, R2RD identifies regime boundaries and partitions the data accordingly. This segmentation ensures that both prediction and interpretation are conditioned on the prevailing regime and provides a structural foundation for regime-aware analysis.

Third, we develop an **attention-based explainability framework** that combines *embedding attention* with *attention rollout*. Embedding attention captures global interaction patterns at the representation level, while attention rollout propagates attention across layers to trace how information flows from inputs to regime-specific predictions. This integrated approach moves beyond raw attention visualization and aligns explanations with the model's internal information routing.

We emphasize **explainability**, demonstrating how attention patterns and feature relevance systematically shift across regimes identified by R2RD. By revealing which signals matter, when they matter, and why, the proposed framework transforms the model from a black-box predictor into a regime-sensitive decision-support tool, bridging machine learning outputs with economic intuition and real-world decision-making.

2 Background

Previous research about XAI focused on testing factor attributions across market regimes, XAI techniques, and model architectures to identify when explanations are stable and trustworthy. Core finance signals, such as Mkt-RF, HML, and CMA, consistently emerge as the top contributors in Transformers and CNNs, while SMB remains weak, aligning with the intuition of asset-pricing.

Bases on this foundation, our work introduces two complementary improvements. First, we expand the dataset to enhance factor coverage and ensure attributions reflect a more representative range of market conditions, increasing robustness and generalizability. Second, we incorporate attention-based visualizations, including rollout and layer-wise inspection, to trace information flow within Transformer layers.

3 Introduction

Machine learning models are increasingly deployed in financial decision-making, ranging from asset pricing and portfolio construction to risk management and regulatory compliance. While these models often deliver strong predictive performance, there are concerns regarding interpretability, robustness, and trust. In finance, explanations are not merely auxiliary diagnostics but integral components of model validation, regulatory disclosure, and economic reasoning. A model whose predictions cannot be economically justified or consistently explained across market conditions is difficult to rely on in practice.

Recent advances in explainable artificial intelligence (XAI) have provided gradient-based attribution methods that assign importance scores to input features based on local sensitivity of model outputs. In financial applications, these methods are frequently used to interpret factor-based models, where attributions can be compared against established asset-pricing intuition. Prior work has benchmarked gradient-based explanations on Transformer architectures using Fama–French factor panels and portfolio returns, highlighting both the promise and the fragility of such explanations, particularly with respect to baseline choice and regime dependence.

3.1 Keywords

3.1.1 Core Concepts

- Explainable AI (XAI): A set of techniques that make complex machine learning models easier to understand by showing why they make certain predictions.
- Attribution: Assigning importance scores to input factors (like financial variables) to explain a model's prediction.

3.1.2 Machine Learning Models

- Convolutional Neural Network (CNN): A type of machine learning model often used for images, here adapted to financial time series. It learns patterns by sliding filters (small windows) across the data.
- Transformer: Another type of model originally designed for language tasks, but also powerful for finance. It learns long-range dependencies in data.
- ReLU (Rectified Linear Unit): A common activation function in neural networks that "turns on" only when input is positive.

3.1.3 XAI Techniques

- Vanilla Gradient (VG): The simplest method—looks at how much the output changes if you nudge an input slightly.
- SmoothGrad (SG): Runs Vanilla Gradient multiple times with small random noise added, then averages results to reduce noise.
- Guided Backprop (GBP): A sharper method that only allows positive signals to flow back, making the explanations cleaner but more fragile.

- SmoothGrad + Guided (SG-GBP): A combination of the above, aiming for both sharpness and stability.
- Integrated Gradients (IG): Instead of looking only at the input, IG compares it to a baseline (like a reference state) and integrates along the path between them to produce more stable attributions.

3.1.4 Integrated Gradients Baselines

- Zero Baseline: Uses "no exposure" as the reference point. (Simple, but unrealistic in finance.)
- Mean Baseline: Uses the average of the dataset or regime as the reference. (Represents a "typical" market state.)
- Random Baseline: Picks a random point from the data as the reference. (Adds variability, can be averaged out.)
- Min/Max Baseline: Uses the lowest or highest observed values. (Stress test extremes, but not always realistic.)
- Extreme Baseline: Uses a manually chosen extreme value (e.g., 3× the mean). (Good for "what-if" stress tests.)

3.1.5 Finance Factors (Fama-French Five-Factor Model)

- Mkt-RF (Market – Risk Free): The excess return of the market compared to a risk-free investment; usually the most important factor
- HML (High Minus Low): Value factor; difference between value stocks (high book-to-market) and growth stocks (low book-to-market)
- SMB (Small Minus Big): Size factor; difference between small and large companies.
- RMW (Robust Minus Weak): Profitability factor; difference between firms with strong vs weak profitability
- CMA (Conservative Minus Aggressive): Investment factor; difference between firms with conservative vs aggressive investment policies.

3.1.6 Regime segmentation

- Regime Detection via RDR2: A distance-based regime segmentation method that identifies structurally similar market states while preserving temporal coherence in non-stationary financial time series.
- Regime-aware Explainability: An interpretability framework that analyzes model explanations conditional on market regimes, enabling comparison of factor importance across different market states.

3.1.7 Attention-Based Explainability

- Attention-based Explainability: Interpretability approaches that use attention weights to understand how neural networks allocate importance across inputs or latent representations.
- Attention Rollout: A visualization method that aggregates attention weights across Transformer layers to trace information flow from inputs to predictions.

3.1.8 Model Interpretability

- Embedding Attention: Attention mechanisms applied at the embedding level, revealing how learned latent representations of input features contribute to model decisions beyond raw input attribution.
- Self-Attention Visualization: The process of visualizing attention weights within self-attention layers to interpret dependencies across features or time steps learned by the model.
- Model Transparency: The degree to which a model's internal decision-making process can be inspected, understood, and communicated to stakeholders, particularly important in high-stakes financial applications.

3.1.9 Conceptual Framing

- Non-stationary Financial Markets: Financial environments in which statistical properties and factor relationships change over time, motivating regime-aware modeling and explainability.
- Explanation Stability: The consistency of interpretability results across models, regimes, or explanation methods, used as a criterion for trustworthiness.
- Mechanism-level Interpretability: An interpretability approach that focuses on internal model mechanisms (e.g., attention routing) rather than solely on output-level sensitivity.

4 Literature review

4.1 XAI in Finance (brief recap)

To interpret model predictions and assess factor relevance, we employ a suite of gradient-based explainability methods that attribute output sensitivity to input features. These methods leverage local derivatives of the model output with respect to the input and are well-suited for continuous, factor-based financial data.

Let $f(\mathbf{x})$ denote a trained model mapping an input vector of factors $\mathbf{x} \in R^d$ to a scalar prediction, such as a portfolio return. Gradient-based attribution methods seek to quantify the contribution of each input dimension x_i to the prediction at a given point $\mathbf{x}$.

Vanilla Gradients and SmoothGrad Vanilla Gradients compute attributions as the partial derivatives of the output with respect to the input,

$$\mathrm{VG}_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}.$$

This measure captures local sensitivity but is known to be noisy and susceptible to gradient saturation.

SmoothGrad mitigates this issue by averaging gradients over noisy perturbations of the input. Specifically, given N samples $\{\boldsymbol{\epsilon}_j\}_{j=1}^N$ drawn from a Gaussian distribution, SmoothGrad computes

$$\mathrm{SG}_i(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^{N} \frac{\partial f(\mathbf{x} + \boldsymbol{\epsilon}_j)}{\partial x_i}.$$

This procedure reduces high-frequency noise and yields more stable attribution maps.

Guided Backpropagation and Guided SmoothGrad Guided Backpropagation modifies the backward pass of the network by suppressing negative gradients at ReLU activations, thereby emphasizing features that positively influence the output. Guided SmoothGrad combines this modification with input perturbation averaging, producing sharper attribution maps that often highlight dominant factors more clearly. However, these guided variants may become unstable in saturated regimes where gradients vanish.

Integrated Gradients Integrated Gradients (IG) addresses gradient saturation by accumulating gradients along a straight-line path from a baseline input $\mathbf{x}'$ to the actual input $\mathbf{x}$. The attribution for feature i is given by

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - x_i') \int_{\alpha=0}^{1} \frac{\partial f\big(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')\big)}{\partial x_i}\, d\alpha.$$

In practice, the integral is approximated using a finite sum. While IG satisfies desirable axioms such as completeness, its attributions depend critically on the choice of baseline. We therefore evaluate multiple baselines, including zero, mean, minimum, maximum, and extreme values, as well as simple multi-baseline averaging, to assess robustness across regimes and architectures.

Aggregation Across Test Samples A key design choice in our analysis is to aggregate attribution maps across multiple test samples rather than relying on explanations from individual observations. Formally, for a given regime, we compute attribution maps for batches of test inputs and report their mean. This aggregation reduces idiosyncratic noise, mitigates path-dependent variability in gradient-based methods, and yields a more stable estimate of global factor importance within each regime. Such averaging is particularly important in financial settings, where single-period observations may be dominated by transient shocks rather than structural relationships.

Together, these gradient-based methods provide complementary perspectives on model behavior. By comparing their outputs across architectures and regime definitions, we assess not only which factors are important, but also how stable and economically meaningful these explanations are.

4.2 Regime Detection in Financial ML

Traditional approaches: k-means, change-point detection (CPD), which often treat observations as i.i.d. and may suffer from over-fragmentation and unstable regime boundaries in non-stationary financial data.

R2RD-style regime segmentation: builds on latent variable models (e.g., Gaussian HMMs) and introduces a rolling, robust framework that emphasizes

- **temporal continuity** through Markovian regime dynamics and rolling retraining,
- **economic interpretability** via regime-level distributional summaries,
- **avoiding fragmentation** using temporal ensemble, label assignment, and threshold policies that stabilize regime identities over time.

Motivation from prior explainability work: earlier findings show that attribution methods such as Integrated Gradients (IG) are highly sensitive to baselines and market conditions, motivating regime-aware evaluation as a necessary layer for robust interpretation.

4.3 Attention Interpretability

Attention explanation and limitation The increasing adoption of Transformer architectures across domains has intensified interest in attention-based interpretability. Because attention mechanisms explicitly compute weighted interactions between input elements, previous work often treated attention weights as direct explanations of model decisions. However, a growing literature has shown that raw attention maps are insufficient, and sometimes misleading, as explanatory tools. This section reviews key findings in attention visualization and motivates the methodological choices in our study.

Previous visualization work utilized attention as a diagnostic tool to better understand model behavior, rather than claiming full causal explanations. Vig (2019) introduces a multiscale visualization framework that exposes attention patterns at different levels of granularity, including individual attention heads, layer-level aggregates, and neuron-level contributions. The framework allows researchers to identify meaningful structural behaviors such as long-range dependencies, syntactic alignment, and head specialization, which are difficult to detect from a single aggregated attention matrix.

A key insight from this work is that attention is inherently multi-dimensional: interpretability depends on which head, which layer, and which aggregation level is examined. While raw attention matrices can reveal relational structure, their interpretive value depends on careful visualization and contextualization. This perspective frames attention primarily as a diagnostic signal, but not necessarily a complete explanation of model decisions.

More recent work has raised the limitation that attention weights alone constitute faithful explanations. Chefer, Gur, and Wolf (2021) provided a systematic critique of attention-based explanations in Transformer models. They argued that raw attention maps ignore several critical components of the architecture, including residual connections, MLP layers, and interactions across attention heads. As a result, attention weights may highlight where the model attends, but not why a particular prediction is produced.

To address these shortcomings, Chefer et al. proposed a relevance propagation framework grounded in Deep Taylor Decomposition. Their method assigns local relevance scores and propagates them backward through attention layers, residual paths, and feed-forward blocks. This ensures conservation of total relevance across layers. They demonstrated that this approach yields sharper and more faithful explanations than both raw attention maps. Their central conclusion, that attention is not explanation, has become a foundational principle in modern Transformer interpretability research.

Together, these studies suggest a nuanced role for attention visualization. On one hand, attention provides valuable insight into how Transformers route information across inputs, especially when examined across layers and heads. On the other hand, attention weights alone do not account for nonlinear transformations, residual mixing, or downstream processing that ultimately drive predictions.

In financial time-series settings, these limitations are particularly salient. Market data are noisy, non-stationary, and highly correlated, which amplifies the risk of over-interpreting raw attention patterns. Consequently, attention visualizations should be treated as mechanistic evidence, rather than as standalone causal explanations. Motivated by this literature, our work adopts attention visualization as a complement to gradient-based attribution methods rather than a replacement. Specifically, we focus on embedding-level attention heatmaps and attention rollout to examine how factor representations interact across regimes, while interpreting these results in conjunction with attribution stability and regime-aware analysis. This combined approach aligns with prior findings by leveraging the strengths of attention visualization while remaining explicit about its limitations as an explanatory tool.

Why attention rollout helps In Transformer models, self-attention is computed layer by layer, where each token aggregates information from other tokens through attention weights. While individual attention maps reveal how tokens attend to each other within a single layer, they do not directly indicate how information from input tokens contributes to the final prediction across multiple layers.

By contrast, Attention Rollout is a heuristic interpretability method that approximates how input token identities propagate through multiple Transformer layers. It assumes that token representations are linearly mixed by attention operations, and therefore recursively composes attention matrices across layers to estimate the cumulative contribution of input tokens to higher-layer representations.

Attention Rollout Formulation. Let N denote the number of input tokens and L the number of Transformer layers. For each layer $l \in \{1, \ldots, L\}$, we denote by $\mathbf{A}^{(l)} \in R^{N \times N}$ the attention matrix, obtained by averaging the attention weights across all heads. Each entry $\mathbf{A}_{ij}^{(l)}$ represents the proportion of information that token i attends to token j at layer l.

To account for residual connections, we define the adjusted attention matrix as

$$\hat{\mathbf{A}}^{(l)} = \alpha \mathbf{A}^{(l)} + (1 - \alpha)\mathbf{I}, \tag{1}$$

where $\mathbf{I}$ is the identity matrix and $\alpha \in (0, 1]$ controls the relative contribution of the attention and residual paths.

The attention rollout matrix $\tilde{\mathbf{A}}^{(L)}$ is computed by recursively composing the adjusted attention matrices across layers:

$$\tilde{\mathbf{A}}^{(L)} = \hat{\mathbf{A}}^{(L)} \hat{\mathbf{A}}^{(L-1)} \cdots \hat{\mathbf{A}}^{(1)}. \tag{2}$$

The resulting matrix $\tilde{\mathbf{A}}^{(L)} \in R^{N \times N}$ encodes the cumulative propagation of token-level information from the input layer to the final layer under the assumption of linear mixing induced by self-attention.

5 Data Preprocessing

5.1 Original Dataset

Our analysis builds on the same dataset used by the previous group, sourced from Kenneth R. French Data Library. Specifically, the dataset combines:

Prediction targets: returns of 10 size-decile–sorted portfolios
Input features: the Fama–French Five Factors, including market excess return (Mkt–RF), size (SMB), value (HML), profitability (RMW), and investment (CMA).

In the original setup, all variables are sampled at a monthly frequency, which yield approximately 750 observations over the full sample period. While this frequency is standard in asset-pricing studies, it shows limitations for regime-based explainability. In particular, when the data are partitioned into multiple regimes, the effective sample size within each regime becomes small, especially for secondary regimes. This reduces the statistical reliability and interpretability of attribution and attention analyses.

5.2 Motivation for Data Expansion

The limited number of observations in the monthly dataset shows a challenge for regime-based explainability. Regime segmentation methods require sufficient data within each regime to support stable model training and meaningful comparison of explanations. With only a few observations in total, regime-specific analyses risk is dominated by sampling noise rather than systematic patterns.

To address this limitation, we expanded the dataset by leveraging higher-frequency data constructed from the same underlying sources and attributes. Our goal is not to change the economic setting, but to increase the effective sample size while preserving consistency with the original factor definitions and portfolio construction.

5.3 Data Expansion Strategy

Instead of relying solely on monthly observations, we construct a daily-frequency dataset covering the same time period and variables. This expansion increases the number of observations to over 20,000 daily data points, substantially enlarging the sample available for regime detection, model training, and explainability analysis.

While the daily dataset offers a much larger sample size, it also comes with high-frequency noise. Daily returns and factor realizations are considerably more volatile and can lead to overfitting, especially for flexible models such as Transformers. Empirically, we observe that models trained directly on daily data tend to capture short-term noise rather than regime-level structure, which result in unstable predictions and explanations.

To balance sample size and noise reduction, we further process the daily data to construct weekly and bi-weekly representations.

5.4 Weekly and Bi-Weekly Aggregation

We generate weekly and bi-weekly datasets using a sliding-window aggregation scheme designed to reduce noise while retaining sufficient temporal resolution for regime learning. We anchor each aggregation window on Wednesdays, which empirically exhibit lower idiosyncratic noise compared with other weekdays and avoid distortions associated with weekend effects or month-end rebalancing.

Sliding windows:

- A 7-day window is used to construct weekly observations.
- A 14-day window is used to construct bi-weekly observations.

Within each window, both factor values and portfolio returns are aggregated using simple averaging, producing smoother time-series while preserving alignment between features and targets.

This approach achieves a trade-off between the stability of lower-frequency data and the richness of higher-frequency sampling. This yields datasets that are both statistically meaningful and suitable for regime-based analysis.

5.5 Benefits for Regime Learning and Explainability

The expanded and aggregated datasets substantially improve the quality of regime-based analysis. First, each regime now contains a sufficiently large number of observations, which enables stable model estimation and more reliable attribution and attention visualization. Second, the smoother weekly and bi-weekly representations allow regimes to capture persistent market states rather than short-lived fluctuations.

6 Methodology

6.1 Model Architecture

6.1.1 Where attention lives

Self-attention in our model resides within the multi-head self-attention modules of each Transformer encoder layer. For each encoder layer, the attention mechanism computes a weighted aggregation of input embeddings, which allows each embedding to selectively incorporate information from other embeddings. Formally, for each attention head, attention weights are computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

where queries Q, keys K, and values V are linear projections of the input embeddings. Each attention head learns a distinct interaction pattern, and multiple heads operate in parallel to capture heterogeneous relationships.

In our implementation, attention weights are extracted per encoder layer and per head, without averaging across heads. This preserves the multi-head structure emphasized in prior interpretability work and avoids collapsing potentially meaningful heterogeneity into a single attention map. The extracted attention tensors have shape:

$$[\text{batch}, \text{heads}, \text{target embeddings}, \text{source embeddings}]$$

which allows us to analyze how each embedding attends to others at different depths of the model.

6.1.2 Embedding structure Factor × Time Representation

Each model input corresponds to a single time step, represented as a vector of financial factors. Rather than treating time as the sequence dimension, we treat factors as tokens, following a factor-as-token formulation commonly used in financial Transformer models.

- Each scalar factor value (e.g., Mkt–RF, HML, RMW, CMA) is independently embedded into a d-dimensional vector using a linear embedding layer.
- The resulting embeddings form a sequence of length equal to the number of factors.
- Positional encoding is added to preserve a consistent ordering across factor embeddings.

This design allows the self-attention mechanism to model cross-factor dependencies, which answers questions such as which factors are most influential when updating the representation of another factor. Importantly, attention is computed over embeddings, not raw input values, meaning that attention weights reflect interactions in latent space after initial feature transformation.

6.2 Regime Segmentation: Gaussian Hidden Markov Model (HMM)

Financial time series are non-stationary: the joint distribution of factor realizations and portfolio returns can shift across market conditions (e.g., calm vs. turbulent periods). To make regime-aware modeling and regime-conditional explainability feasible, we segment the sample into a small number of persistent "market states" using a Gaussian Hidden Markov Model (HMM), which explicitly models temporal persistence via a Markov transition structure rather than treating time points as i.i.d. clusters.

6.2.1 Model Setup

Let $x_t \in R^d$ denote the observed factor vector at time t (e.g., Fama–French factors). The HMM assumes an unobserved discrete regime variable $z_t \in \{1, \ldots, K\}$ such that:

- **State dynamics (temporal persistence):**

$$\Pr(z_t = j \mid z_{t-1} = i) = A_{ij}, \qquad \Pr(z_1 = j) = \pi_j,$$

  where A is the $K \times K$ transition matrix and π is the initial state distribution.

- **Gaussian emissions (state-conditional factor distribution):**

$$x_t \mid (z_t = k) \sim \mathcal{N}(\mu_k, \Sigma_k).$$

  We use a full-covariance Gaussian emission model, allowing correlated factor movements within each regime.

This formulation yields regimes that are (i) structurally distinct in factor space via (μ_k, Σ_k) and (ii) temporally coherent via A, which is important for regime-aware slicing and comparisons in explainability.

6.2.2 Preprocessing and Observation Matrix

We construct the observation matrix $X = [x_1, \ldots, x_T]^\top$ from the factor dataset and standardize each factor to zero mean and unit variance:

$$\tilde{x}_t = \frac{x_t - \bar{x}}{s_x}.$$

In practice, we apply `StandardScaler()` and feed the standardized factor matrix into the HMM.

6.2.3 Selecting the Number of Regimes K

The choice of the number of regimes K is an important modeling decision. In principle, model selection criteria such as information criteria (e.g., Bayesian Information Criterion, BIC) or clustering-based separation metrics (e.g., silhouette score) can be used to assess candidate values of K.

In practice, we compute these diagnostics to inform the choice of K, but we do not mechanically select K by optimizing a single criterion. Instead, the final value of K is fixed to a small number based on a combination of diagnostic evidence, domain knowledge, and empirical sanity checks. Specifically, we choose K such that:

- **Regimes are sufficiently populated**, avoiding trivially small states that would lead to unstable downstream estimation;
- **Temporal persistence is preserved**, with regimes forming coherent contiguous segments consistent with the Markov assumption;
- **Economic interpretability is maintained**, allowing regimes to be meaningfully summarized and compared in terms of factor behavior and market conditions.

This approach treats information criteria and separation metrics as diagnostic tools rather than strict optimization objectives. By fixing K at a parsimonious level, we prioritize the stability and interpretability of regime-conditioned modeling and explainability over fine-grained likelihood-based optimization, which is sufficient for our goal of regime-aware analysis rather than precise regime discovery.

6.2.4 Fitting and Decoding Regimes

For the selected K, we fit a final Gaussian HMM and decode regime labels using the most likely state sequence (Viterbi decoding via `predict`). The decoded labels are stored as:

$$\texttt{df["Regime"]} \leftarrow \hat{z}_{1:T}.$$

We also report per-regime sample counts as a basic sanity check to ensure regimes are sufficiently populated for downstream training and explainability analysis.

6.2.5 How Regimes Are Used Downstream

The HMM regimes are used in two ways:

- **Regime-aware modeling:** train/evaluate models within each regime (or compare regime-conditioned performance) to reduce non-stationarity-induced instability.
- **Regime-conditioned explainability:** compute and compare attention visualizations / rollout / attribution statistics within each regime to test whether "what the model attends to" is stable or shifts across market states.

This design supports apples-to-apples comparisons of explanations within relatively homogeneous market conditions.

6.2.6 Practical Diagnostics and Caveats

To ensure regimes are meaningful (not artifacts), we recommend reporting:

- **Regime size balance:** no regime should be trivially small; otherwise regime-specific estimation becomes unreliable.
- **Temporal coherence:** regimes should form persistent blocks rather than rapidly switching every period (consistent with the Markov assumption).
- **Economic interpretability:** summarize each regime via simple statistics (e.g., mean/volatility of returns, factor means/variances, correlation patterns) to link regimes to market narratives rather than purely statistical partitions.

Finally, note that Gaussian HMM regimes are descriptive rather than causal: they capture recurring patterns in factor distributions and their persistence, but they do not by themselves explain why a regime occurs.

6.3 Explainability Methods

6.3.1 Gradient-based

To interpret model predictions and assess factor relevance, we employ a suite of gradient-based explainability methods that attribute output sensitivity to input features. These methods leverage local derivatives of the model output with respect to the input and are well-suited for continuous, factor-based financial data.

Let $f(\mathbf{x})$ denote a trained model mapping an input vector of factors $\mathbf{x} \in R^d$ to a scalar prediction, such as a portfolio return. Gradient-based attribution methods seek to quantify the contribution of each input dimension x_i to the prediction at a given point $\mathbf{x}$.

Vanilla Gradients and SmoothGrad Vanilla Gradients compute attributions as the partial derivatives of the output with respect to the input,

$$\mathrm{VG}_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}.$$

This measure captures local sensitivity but is known to be noisy and susceptible to gradient saturation.

SmoothGrad mitigates this issue by averaging gradients over noisy perturbations of the input. Specifically, given N samples $\{\boldsymbol{\epsilon}_j\}_{j=1}^N$ drawn from a Gaussian distribution, SmoothGrad computes

$$\mathrm{SG}_i(\mathbf{x}) = \frac{1}{N}\sum_{j=1}^{N} \frac{\partial f(\mathbf{x}+\boldsymbol{\epsilon}_j)}{\partial x_i}.$$

This procedure reduces high-frequency noise and yields more stable attribution maps.

Guided Backpropagation and Guided SmoothGrad Guided Backpropagation modifies the backward pass of the network by suppressing negative gradients at ReLU activations, thereby emphasizing features that positively influence the output. Guided SmoothGrad combines this modification with input perturbation averaging, producing sharper attribution maps that often highlight dominant factors more clearly. However, these guided variants may become unstable in saturated regimes where gradients vanish.

Integrated Gradients Integrated Gradients (IG) addresses gradient saturation by accumulating gradients along a straight-line path from a baseline input $\mathbf{x}'$ to the actual input $\mathbf{x}$. The attribution for feature i is given by

$$\mathrm{IG}_i(\mathbf{x}) = (x_i - x_i') \int_{\alpha=0}^{1} \frac{\partial f\big(\mathbf{x}' + \alpha(\mathbf{x}-\mathbf{x}')\big)}{\partial x_i}\, d\alpha.$$

In practice, the integral is approximated using a finite sum. While IG satisfies desirable axioms such as completeness, its attributions depend critically on the choice of baseline. We therefore evaluate multiple baselines, including zero, mean, minimum, maximum, and extreme values, as well as simple multi-baseline averaging, to assess robustness across regimes and architectures.

Aggregation Across Test Samples A key design choice in our analysis is to aggregate attribution maps across multiple test samples rather than relying on explanations from individual observations. Formally, for a given regime, we compute attribution maps for batches of test inputs and report their mean. This aggregation reduces idiosyncratic noise, mitigates path-dependent variability in gradient-based methods, and yields a more stable estimate of global factor importance within each regime. Such averaging is particularly important in financial settings, where single-period observations may be dominated by transient shocks rather than structural relationships.

Together, these gradient-based methods provide complementary perspectives on model behavior. By comparing their outputs across architectures and regime definitions, we assess not only which factors are important, but also how stable and economically meaningful these explanations are.

6.3.2 Attention-based

We define embedding attention as the attention computed between factor embeddings within the Transformer encoder. Each attention matrix therefore represents how the embedding of a target factor aggregates information from source factor embeddings.

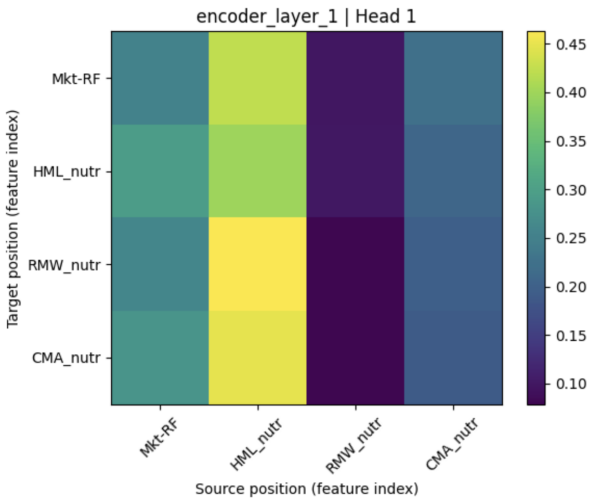

Figure 1: Attention Heatmap

In attention heatmaps:

- Rows (targets) correspond to embeddings being updated,
- Columns (sources) correspond to embeddings providing information,
- Brighter values indicate stronger reliance of the target embedding on the source embedding.

This representation enables direct visualization of learned **c**ross-factor interaction structure, rather than merely highlighting input importance in isolation.

7 Experiments

7.1 Experimental Design

Why Aggregate XAI over Intervals Instead of Single Sample

Original Method: Single-Sample XAI In the original implementation, the XAI gradients (Vanilla Gradient, Integrated Gradients, SmoothGrad, etc.) were computed for a **single test sample** x_i:

$$\text{XAI}(x_i) = \nabla_x f_\theta(x_i) \in R^{d_{\text{features}}}$$

where f_θ is the trained model and $\nabla_x f_\theta(x_i)$ measures the sensitivity of the output with respect to each input feature.

The gradient for a single sample may be highly influenced by noise or outliers, and may not represent the overall feature importance for the regime or the test set. For example, in financial time series, different timestamps may exhibit drastically different volatility, leading to highly variable gradients.

Interval Averaging To obtain more stable and reliable XAI attributions, we can average the gradients over fixed intervals of samples (for example, every 5 samples):

$$\text{XAI}_{\text{interval}} = \frac{1}{n_{\text{interval}}} \sum_{i \in \text{interval}} \nabla_x f_\theta(x_i)$$

Then, we take the mean across all intervals in the test set:

$$\text{XAI}_{\text{mean}} = \frac{1}{N_{\text{intervals}}} \sum_{\text{interval}} \text{XAI}_{\text{interval}}$$

where

$$N_{\text{intervals}} = \left\lceil \frac{N_{\text{test}}}{\text{interval}} \right\rceil, \quad n_{\text{interval}} \leq \text{interval}$$

The resulting $\text{XAI}_{\text{mean}} \in R^{d_{\text{features}}}$ represents the average contribution of each feature over the test set.

1. **Reduce Noise Influence:** Single-sample gradients can be affected by random fluctuations or extreme points. Interval averaging acts as a local smoothing, providing a more reliable estimate of feature importance.

2. **Capture Collective Feature Behavior:** A regime may contain samples from different market states (e.g., high volatility, low volatility). Single-sample XAI cannot capture general patterns, while interval averaging reflects typical behavior within a regime.

3. **Stable Visualization:** Plotting single-sample gradients often produces volatile or noisy heatmaps/bar plots, making interpretation difficult. Averaged gradients are smoother and easier to interpret.

This XAI_{mean} provides a more accurate reflection of the average impact of each feature on the predictions across the entire test set or regime.

8 Results

8.1 From Raw to Embedding-Level Attention

Before introducing attention rollout, we briefly discuss the limitations of raw, single-layer attention extracted directly from the Transformer encoder. While such attention maps are often interpreted as indicators of feature importance, they exhibit substantial instability across heads and layers, limiting their explanatory reliability.

To address this issue, we aggregate attention across heads and project it to the embedding level. All attribution results presented in the following subsections are therefore based on embedding-level attention, rather than raw single-head attention.

Attention Attribution Patterns
Across regimes, raw attention maps exhibit the following high-level attribution tendencies:

- **Mkt–RF** consistently receives strong attention from most other factors, indicating its central role in the learned representations.
- **RMW** and **HML** typically receive moderate levels of attention.
- **CMA** generally receives weaker attention relative to the other factors.

While these patterns align loosely with economic intuition, they are not stable across layers or regimes, raising concerns about the robustness of raw attention as an explanatory signal.

Robustness Analysis
A closer inspection reveals several sources of instability in the raw attention heatmaps:

- The dominant factor frequently switches across regimes, with CMA appearing dominant in some regimes and Mkt–RF in others, even when the underlying market conditions are similar.
- The relative ordering of HML, RMW, and CMA varies across both layers and regimes, suggesting sensitivity to architectural depth rather than structural economic relationships.
- Self-attention strength fluctuates substantially, sometimes appearing weak and other times dominant, which indicates layer-specific noise rather than consistent dependence patterns.

These findings indicate that while embedding-level attention improves interpretability relative to raw attention, it does not provide a stable or global explanation of factor importance.

Embedding-level attention remains a local representation, reflecting information flow within a single layer or limited depth of the model. It does not distinguish between direct and indirect dependencies, nor does it account for cumulative information propagation across stacked Transformer layers.

These limitations motivate the use of attention rollout, which aggregates attention across layers to recover global information pathways and provide a more stable and holistic interpretation of factor interactions.

8.2 Attribution vs Attention: Consistency

In this analysis, we examine the extent to which gradient-based attribution methods and attention-based interpretations agree in identifying important economic factors.

Analysis results indicate a strong consistency between attribution and attention in highlighting the primary economic drivers. In both Regime 0 and Regime 1, IG assigns the largest attribution scores to a small set of factors, notably the market excess return (*Mkt–RF*) and, to a lesser extent, size or profitability factors.

These factors account for the majority of the predicted output variation, suggesting the model heavily relies on core economic drivers. Correspondingly, attention rollout heatmaps consistently show these same factors as dominant sources across layers, indicating that information originating from these factors is propagated extensively throughout the model.

The observed alignment arises from the complementary nature of the two perspectives. While IG captures the integrated effect of a factor along a baseline-to-input trajectory, quantifying its direct contribution to the model output, attention rollout, traces how information flows internally through the network, includinginteractions between factors that may not have immediate marginal effects on the output.

When the attribution is dominated by key economic factors, the attention rollout propagation pattern naturally emphasizes the same factors, leading to an overall agreement.

From the economical perspective, this alignment reinforces the central role of fundamental risk factors in shaping the model's predictions. It confirms that factors such as *Mkt–RF* serve both as primary drivers of output sensitivity (as measured by IG) and as key conduits of information within the Transformer (as measured by rollout), highlighting their dual importance in both predictive performance and internal representation.

In summary, while attention and attribution differ in scope and methodology, they converge in regimes where a few dominant economic factors govern the model's output.

8.3 Attention Rollout for Factor Interaction Analysis

Extraction of Attention Weights. During the forward pass, attention weights from each self-attention layer are stored and extracted for interpretability analysis. For each Transformer layer l, the raw attention tensor has the shape (batch $\times$ heads, N, N), where N denotes the number of input factors. These attention weights describe how each factor (as a target) attends to other factors (as sources) when forming its latent representation.

Head Aggregation and Normalization. To obtain a single attention matrix per layer, attention weights are averaged across all heads, yielding

$$\mathbf{A}^{(l)} \in R^{N\times N}.$$

Each entry $\mathbf{A}_{ij}^{(l)}$ represents the relative contribution of factor j to the updated representation of factor i at layer l. The attention matrix is then normalized row-wise so that the weights for each target factor sum to one. From an economic perspective, this normalization implies that the information used to update a given factor is decomposed into weighted contributions from all other factors, including the market factor and style factors.

Attention Rollout Across Layers. To capture how information propagates across multiple Transformer layers, attention rollout is computed by recursively composing attention matrices:

$$\tilde{\mathbf{A}}^{(L)} = \mathbf{A}^{(1)}\mathbf{A}^{(2)}\cdots\mathbf{A}^{(L)},$$

where L denotes the total number of Transformer layers. The resulting matrix $\tilde{\mathbf{A}}^{(L)} \in R^{N\times N}$ encodes the cumulative propagation of factor-level information from the input layer to the final layer under the assumption of linear mixing induced by self-attention.

Economic Interpretation of the Rollout Matrix. The final attention rollout matrix provides a global view of the factor dependencies learned by the model. A large value in $\tilde{\mathbf{A}}_{ij}^{(L)}$ indicates that factor j (e.g., market excess return) exerts a strong cumulative influence on factor i (e.g., profitability or investment) through multiple layers. Diagonal entries reflect the persistence of a factor's own information across layers, while strong off-diagonal entries reveal cross-factor transmission channels, such as market conditions shaping size, value, or investment dynamics.

Overall, attention rollout aggregates layer-wise attention into a single, interpretable matrix that approximates how economically meaningful risk factors interact and propagate through the Transformer. Rather than focusing on individual heads or layers, this approach provides a holistic visualization of information flow, highlighting dominant economic drivers and their interdependencies in the learned representation.

8.4 Attention Rollout Visualizations

The attention rollout heatmaps exhibit a high degree of consistency across Regime 0, Regime 1, and Regime 2. In all three regimes, the market factor (*Mkt–RF*) consistently appears as a dominant source factor, exerting a strong influence on the representations of all other factors. Quantitatively, the rollout weights associated with the market factor as the source position exceed 0.25 for every target factor, indicating a substantial and persistent contribution.

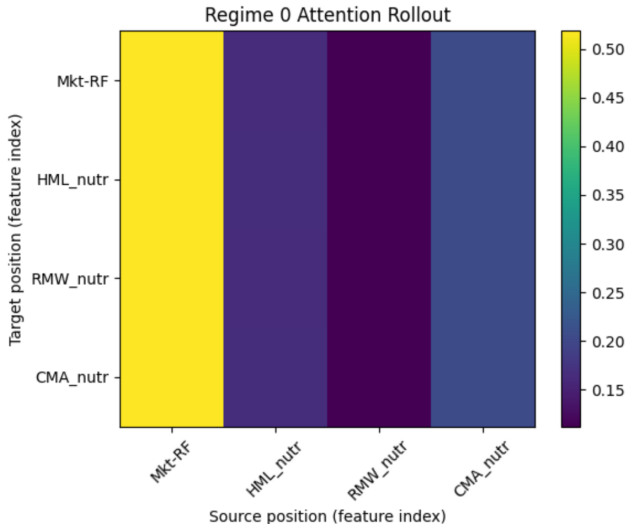

Figure 2: Attention Heatmap

The attention rollout visualization provides a mechanism to quantify the effective contribution of each input feature to the model's predictions after propagating through all layers of the transformer. By examining the rollout maps across different regimes, it becomes apparent that the feature **Mkt-RF** consistently receives a substantial share of the attention weight. This pattern is observed not only in a single regime but also across multiple regimes, suggesting that the model relies on Mkt-RF information more heavily and consistently than on other features such as HML, RMW, or CMA.

The persistent prominence of Mkt-RF in the attention maps implies that it serves as a globally important signal for the prediction task. Even as the underlying market conditions change, reflected in different regimes, the attention attributed to Mkt-RF remains relatively stable. This robustness indicates that the model identifies Mkt-RF as a fundamental driver of the target variable, capturing information that is broadly relevant across various market states. In other words, Mkt-RF provides a feature representation that consistently informs the model's outputs regardless of regime shifts.

From the perspective of explainable AI, the attention rollout captures how information flows and is weighted across the network layers. The fact that Mkt-RF retains a high weight through this propagation suggests that the model prioritizes it over other features, highlighting its predictive importance. The robustness of this pattern across regimes further reinforces the interpretation that Mkt-RF encodes a stable and globally significant factor in the dataset. Consequently, Mkt-RF can be considered a core feature whose variations the model systematically leverages to produce accurate predictions, demonstrating both interpretability and consistency in the learned representations.

9 Discussion

Economic Interpretation. This finding aligns closely with the results obtained from the previous factor group analysis, which also concluded that the market factor plays a central role in shaping the model's predictions. From an economic perspective, the dominance of the market factor suggests that aggregate market risk serves as a primary channel through which information is transmitted to style factors such as size, value, profitability, and investment, across different market regimes.

Advantage of Rollout-Based Visualization. Compared to single-layer attention visualizations, attention rollout provides a more comprehensive representation of factor influence by explicitly accounting for information propagation across multiple Transformer layers. By aggregating attention weights across the entire network, the rollout visualization captures the cumulative effect of inter-factor interactions, rather than isolated, layer-specific dependencies. As a result, it offers a more faithful depiction of how economically meaningful factors influence the model as a whole, explaining both the observed robustness across regimes and the persistent dominance of the market factor.

10 Limitations and Future Work

10.1 Expand the Explainability Technique Beyond Gradient-based

While Integrated Gradients (IG) provides a principled gradient-based attribution method with axiomatic guarantees, relying on a single class of explainability techniques may offer an incomplete view of a model's decision process. To broaden the scope of interpretability, we can consider Shapley Additive Explanations (SHAP), a class of attribution methods grounded in cooperative game theory.

SHAP explains a model's prediction by decomposing the output into additive contributions from each input feature, where each contribution corresponds to the feature's Shapley value. In the context of factor-based return prediction, SHAP quantifies how much each factor contributes, on average, to increasing or decreasing the model's predicted return relative to a reference state. This formulation aligns naturally with the economic interpretation of factor models, in which portfolio returns are understood as the aggregation of marginal factor effects.

A key component of SHAP is the specification of a *background distribution*, also referred to as the baseline dataset. The background distribution represents the model's normal or reference operating environment and serves as a proxy for the absence of informative signal. Feature attributions are then computed by comparing the model's output when a given factor is present to its expected output under this background distribution. In essence, SHAP measures how the inclusion of a factor shifts the prediction away from a baseline expectation.

Compared to gradient-based methods, SHAP offers several complementary advantages. First, it is model-agnostic in principle, enabling consistent comparisons across architectures such as Transformers and convolutional neural networks. Second, SHAP explicitly accounts for nonlinear interactions among factors by averaging marginal contributions over all possible feature coalitions.

Agreement across different methods will strengthen confidence in the inferred economic drivers, while systematic divergence may reveal nonlinear interactions, regime dependence, or structural properties of the model that warrant further investigation.

10.2 Applying Attention Flow

While attention rollout provides a useful first-order approximation of how information propagates across Transformer layers, it remains a linear and unconstrained aggregation of attention matrices. In future work, a natural extension of our framework is to apply *attention flow*, which models attention propagation as a structured flow problem rather than a simple matrix product.

Attention rollout implicitly assumes that attention weights across layers can be composed multiplicatively, treating each layer as an independent transition matrix. As a result, rollout captures average influence but

may obscure how attribution is routed through competing paths in deeper architectures.

Attention flow addresses these limitations by interpreting attention as a directed weighted graph and computing relevance as a flow that must satisfy conservation constraints. This flow-based perspective makes attribution path-sensitive and explicitly accounts for bottlenecks, dominance, and diversion of attention across layers. Attention flow naturally aligns with regime-based analysis. By computing flow-based attributions separately within each regime, one can study how the structure of information routing changes across market states, rather than only comparing marginal importance levels. Persistent flow patterns across regimes would indicate structural reliance on certain factors, whereas regime-dependent re-routing would suggest adaptive or conditional factor usage by the model.

Future work could examine whether high-flow paths correspond to stable gradient attributions and whether discrepancies between the two signal nonlinear interactions or saturation effects. Such cross-method validation would further strengthen the interpretability and trustworthiness of deep learning models in asset pricing.

11 Conclusion

This study applies a Gaussian Hidden Markov Model (HMM) to standardized multi-factor data for market regime segmentation, with the number of regimes selected using Silhouette scores and BIC. The results show that the model captures distinct and temporally coherent market states, providing a robust and interpretable foundation for downstream tasks such as risk management, strategy switching, and market scenario analysis.

12 Reference

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Vig, J. (2019). A Multiscale Visualization of Attention in the Transformer Model. arXiv preprint arXiv:1906.05714. https://doi.org/10.48550/arXiv.1906.05714

For Data:

Prediction target: 10 size dencile sorted portfolios
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Features: Fama-French 5 (FF5) factors. Data description is in
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_5_factors_2x3.html

13 Appendix

```
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Rows: 1189
Epoch 2/10  Loss=0.0241
Epoch 4/10  Loss=0.0071
Epoch 6/10  Loss=0.0041
Epoch 8/10  Loss=0.0034
Epoch 10/10  Loss=0.0032
Regime 0 RMSE: 0.0278
XAI for Regime 0, mean over test set every 5 samples:
Vanilla Gradient: [ 0.19252732  0.01172869  0.00536786 -0.01336746]
Integrated Gradients: [ 1.1749006e-02 -4.6049303e-04  2.9680732e-05  9.4337791e-04]
SmoothGrad (vanilla): [ 0.19256513  0.01173866  0.00537144 -0.01338399]
Guided Backprop: [ 0.18756506  0.01047815  0.00468917 -0.00718996]
SmoothGrad + Guided: [ 0.18751143  0.01047712  0.0046944  -0.00721057]
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Layer 1 raw attn shape: (1, 4, 4, 4)
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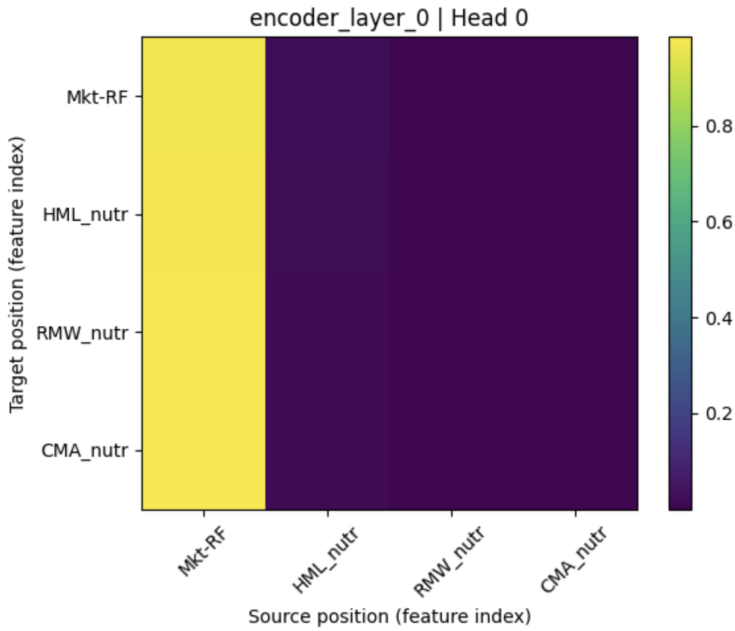

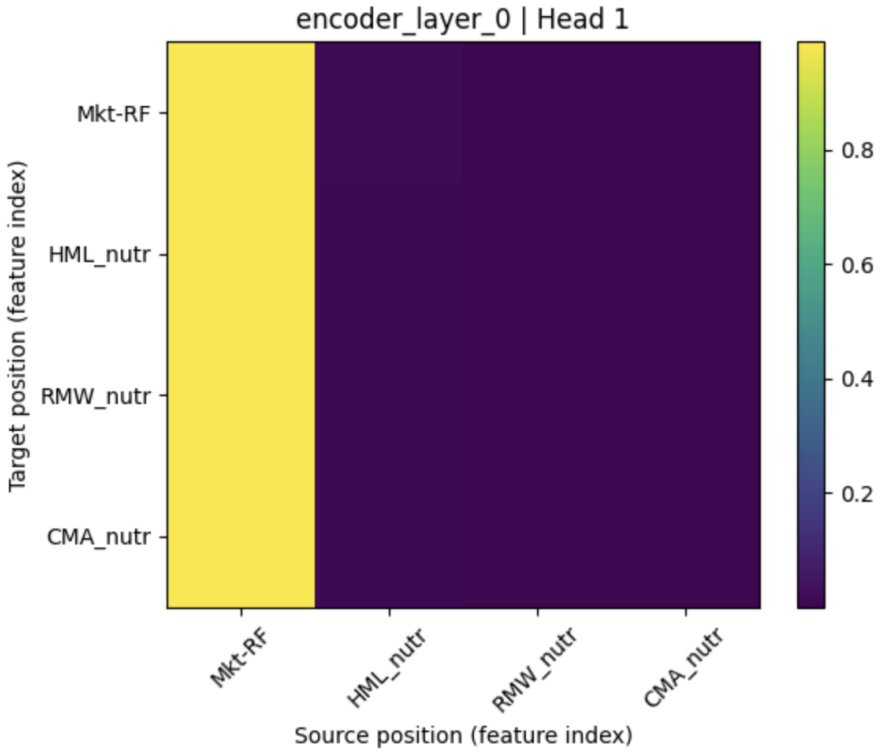
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CMA_nutr
Source position (feature index)
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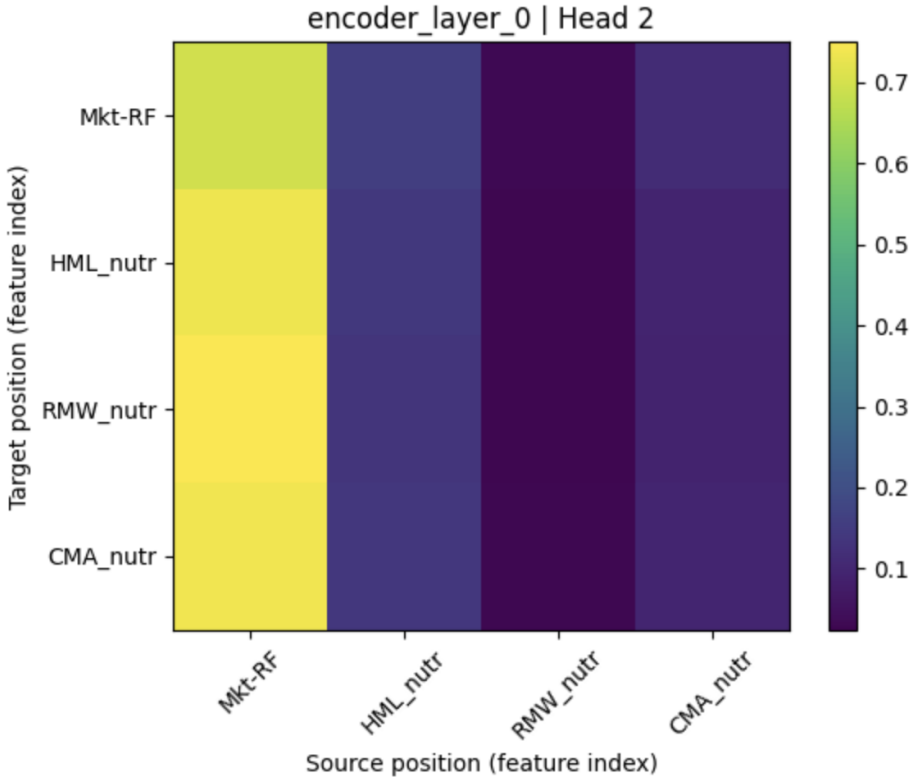
0.4
0.2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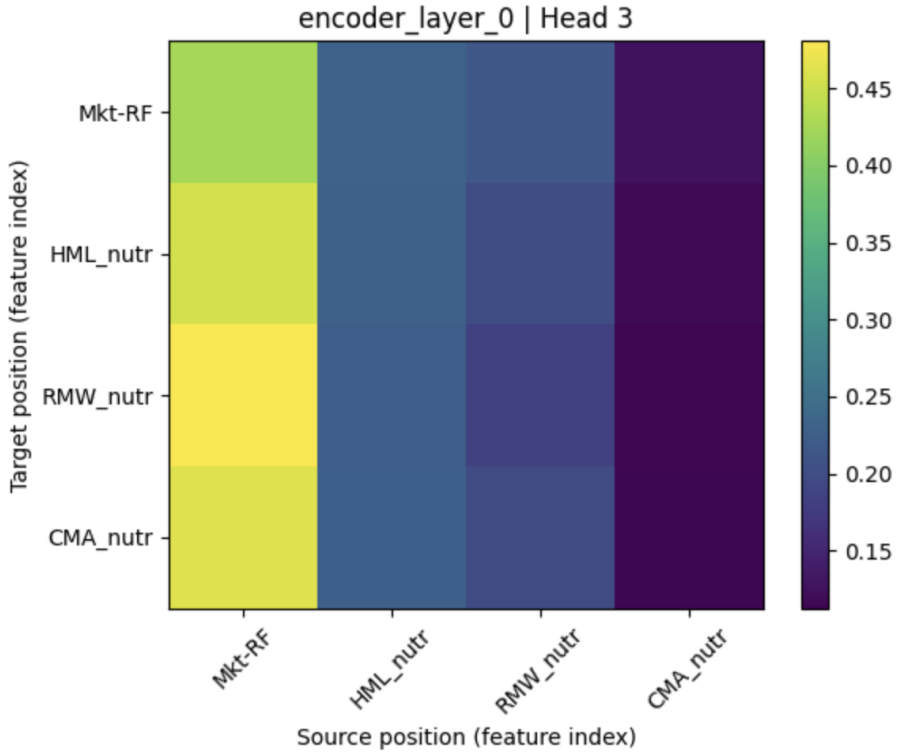
0.2
0.1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CMA_nutr
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Source position (feature index)
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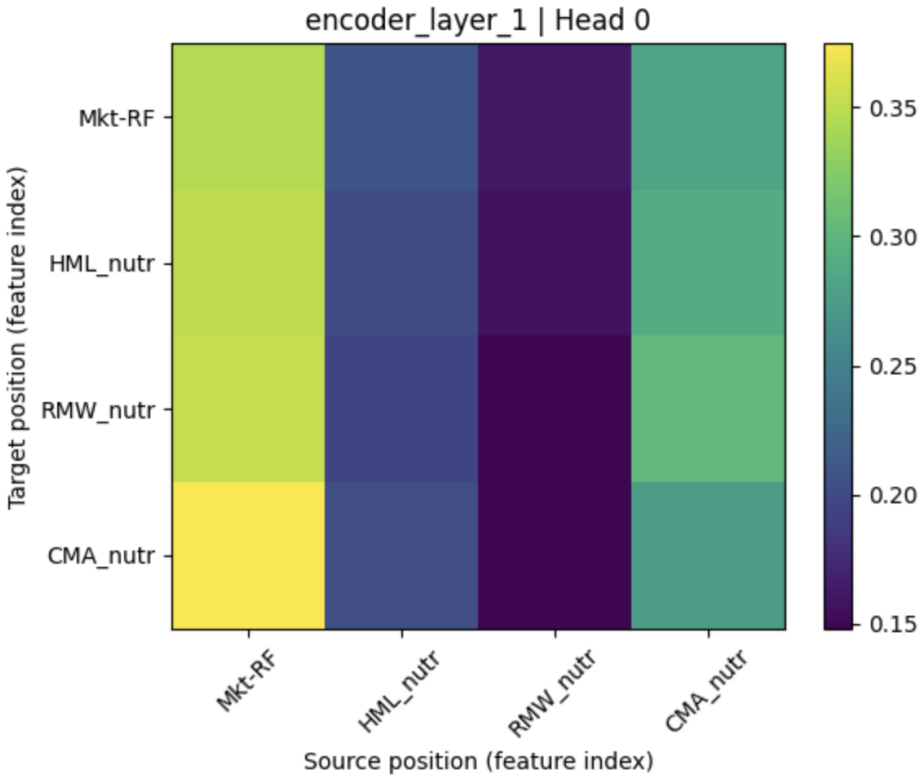
0.15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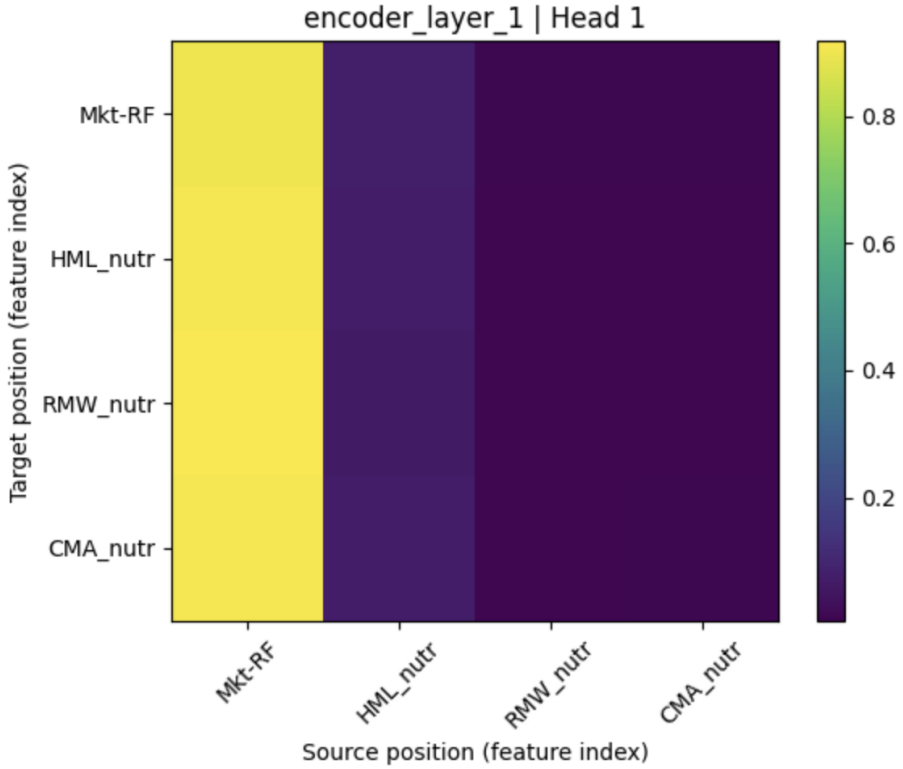
0.20
0.15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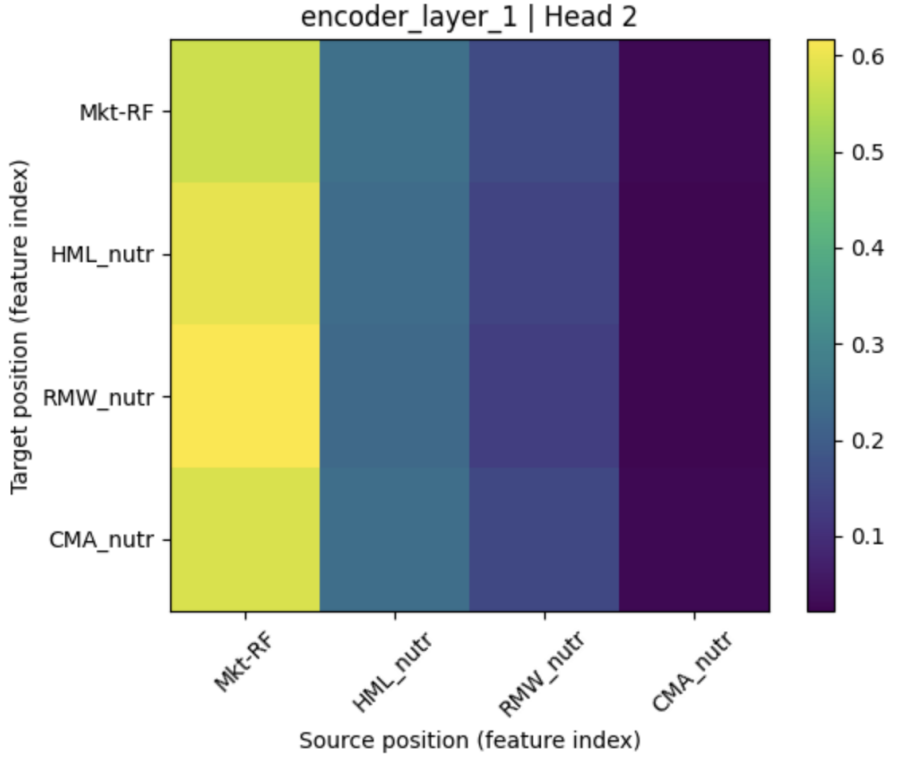
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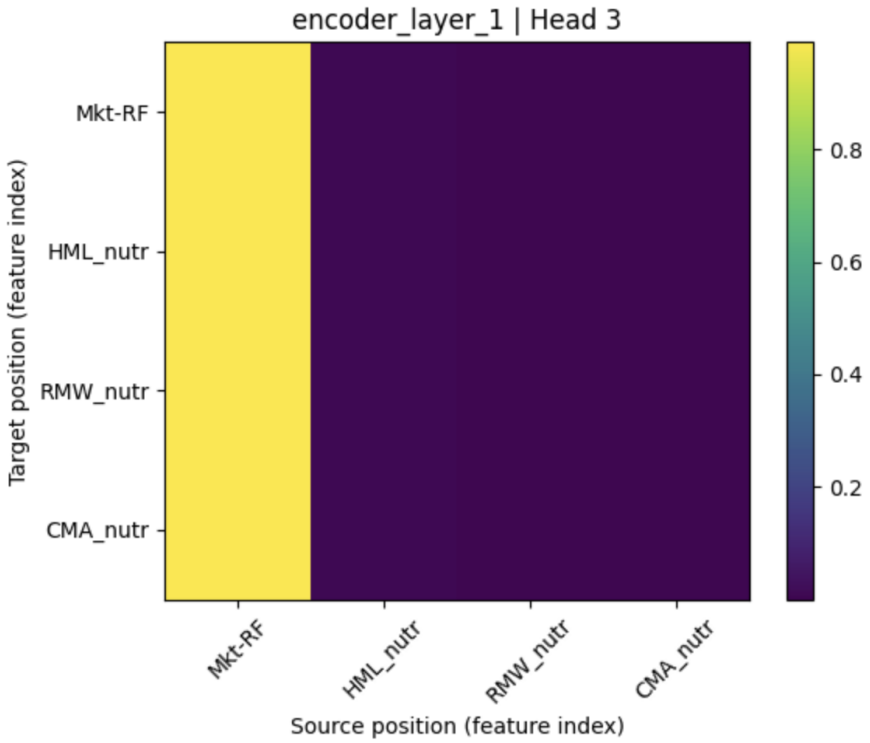
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0.2

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Epoch 4/10  Loss=0.0456
Epoch 6/10  Loss=0.0219
Epoch 8/10  Loss=0.0195
Epoch 10/10  Loss=0.0176
Regime 1 RMSE: 0.1007
XAI for Regime 1, mean over test set every 5 samples:
Vanilla Gradient: [ 0.19155484  0.01526179 -0.00230889 -0.01341872]
Integrated Gradients: [ 0.00812887 -0.00027246  0.0001489  -0.00253215]
SmoothGrad (vanilla): [ 0.19143118  0.01525848 -0.0023162  -0.01332386]
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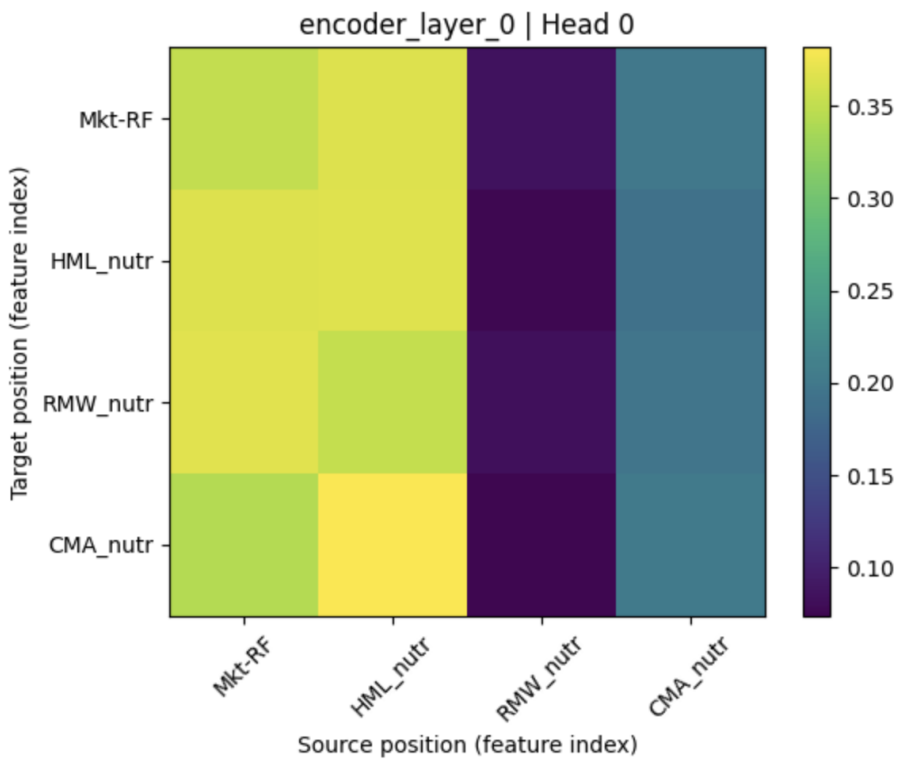

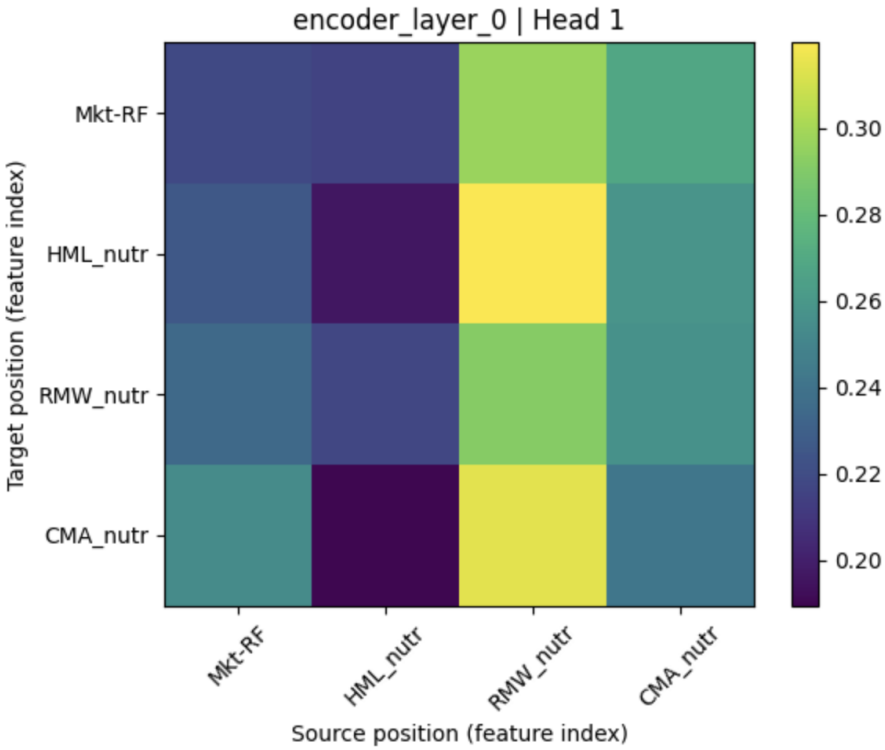
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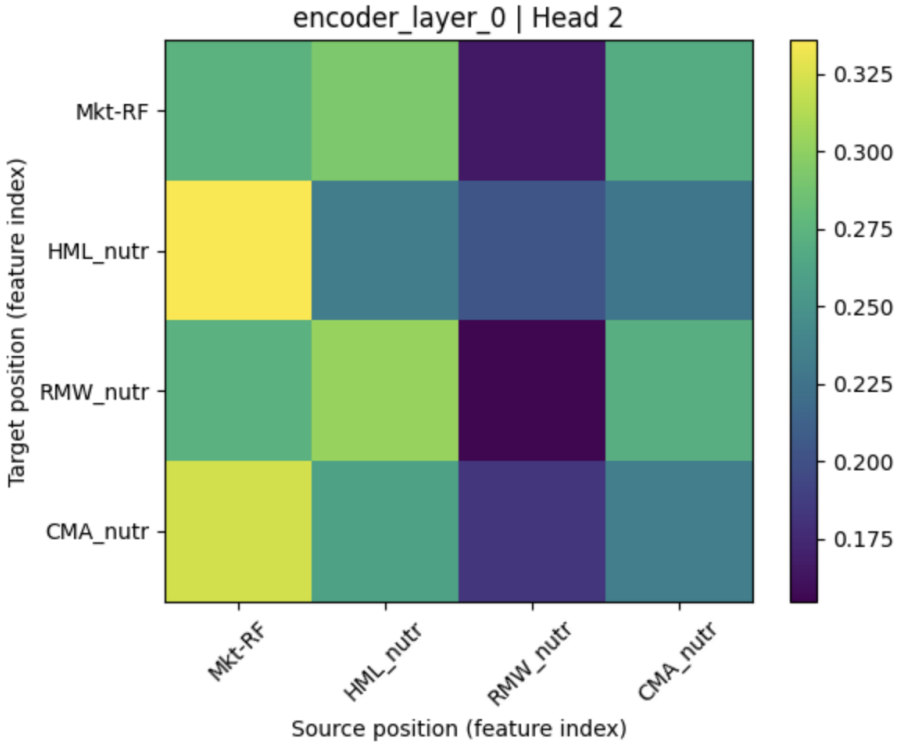
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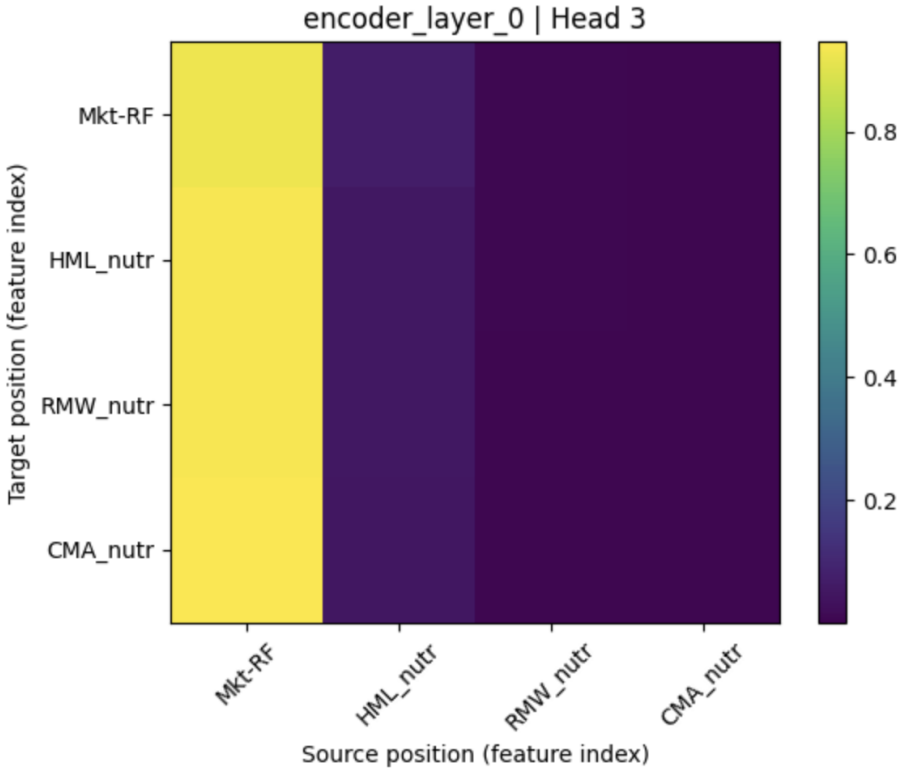
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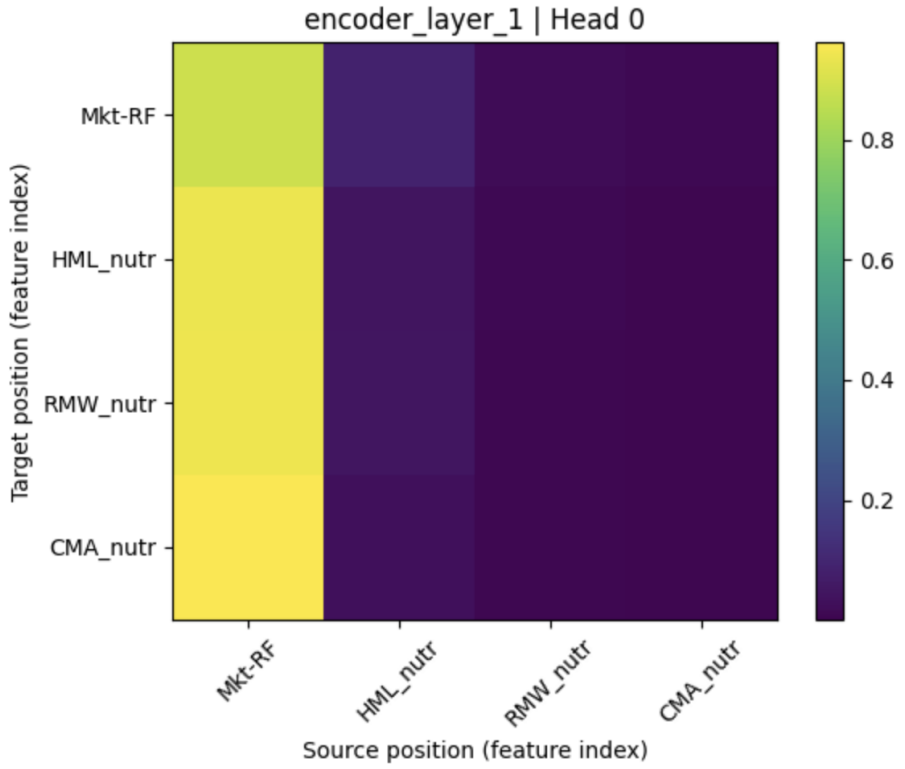
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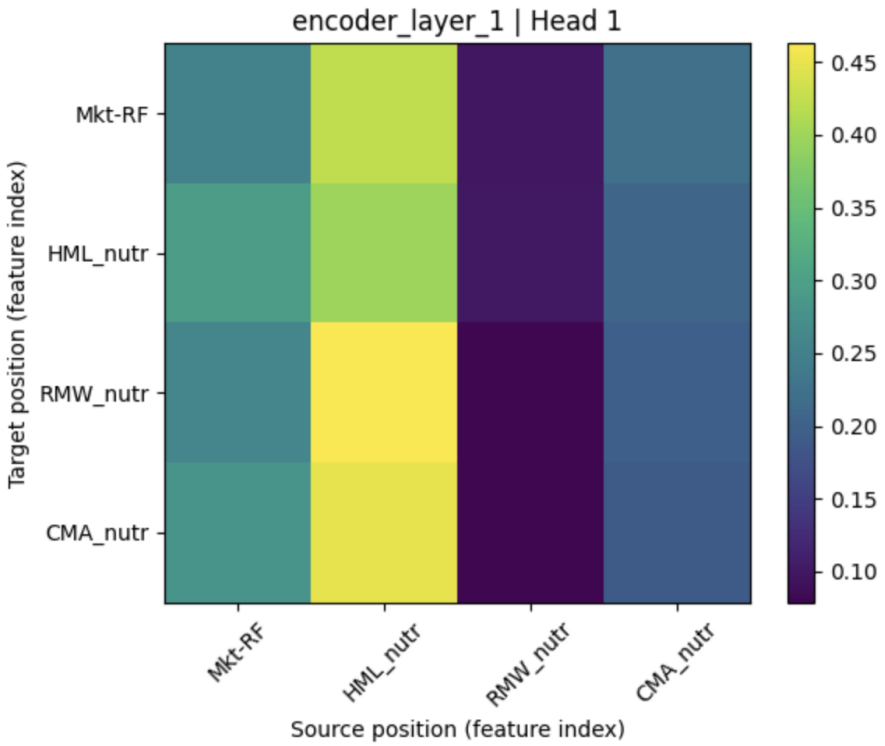
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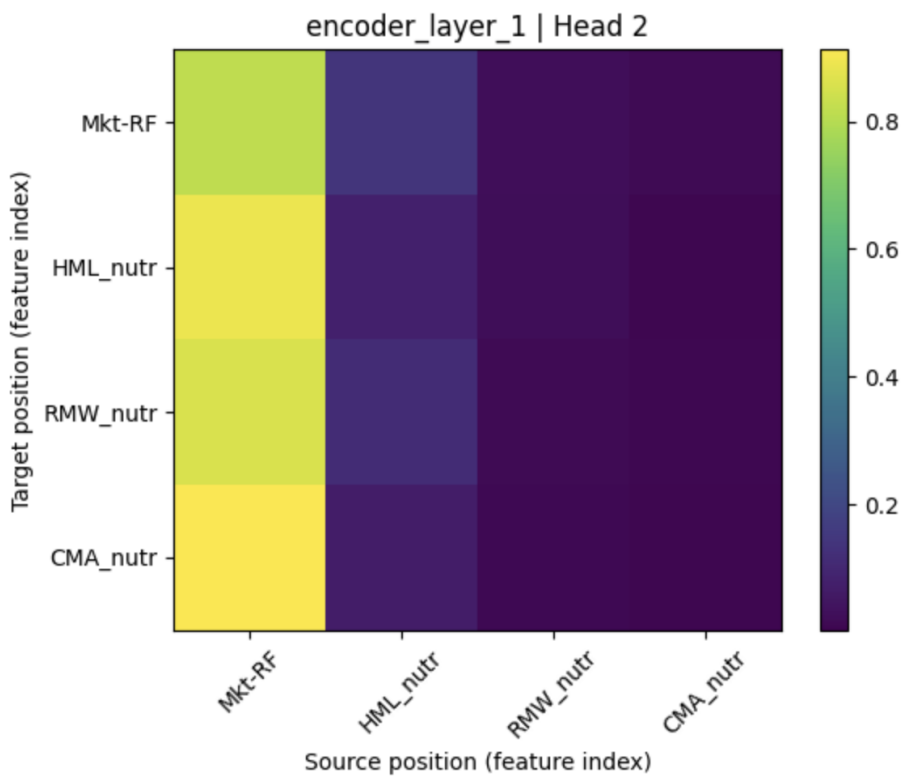
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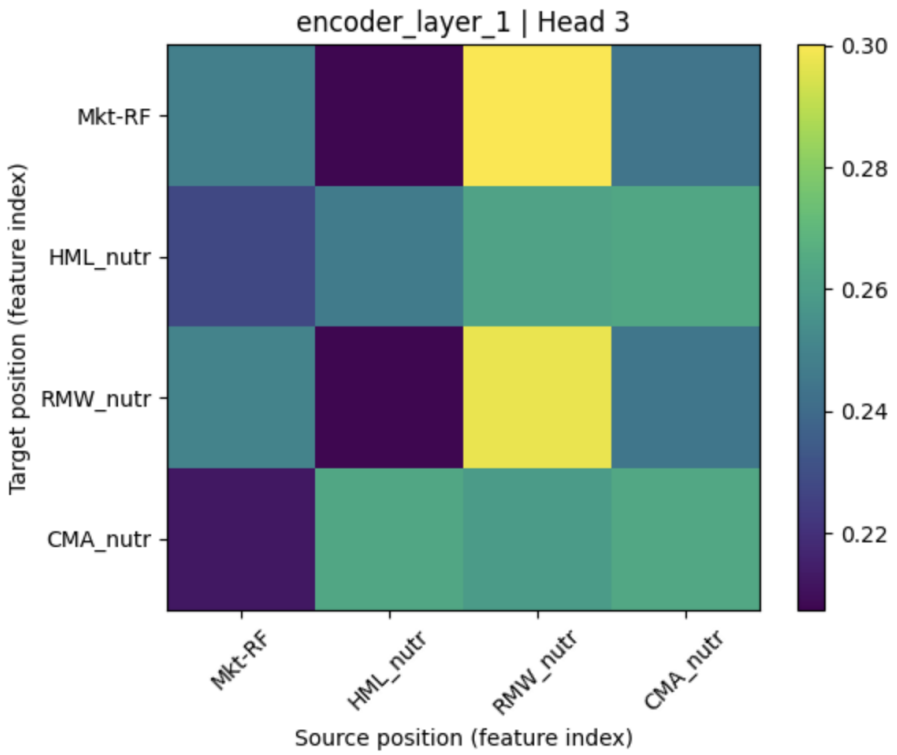
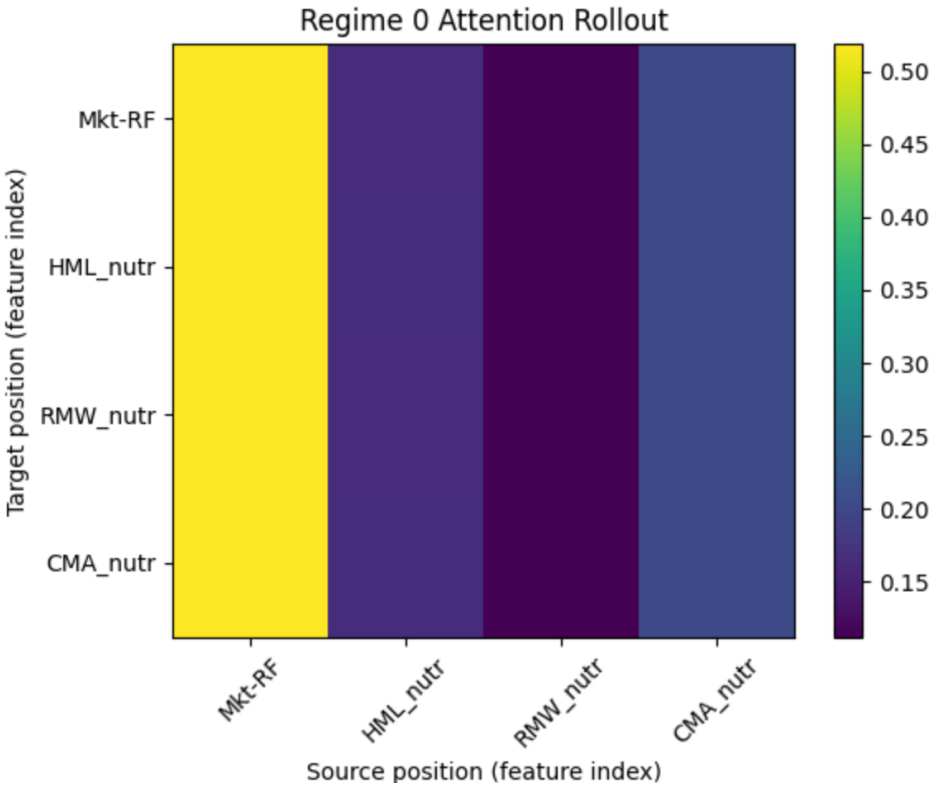
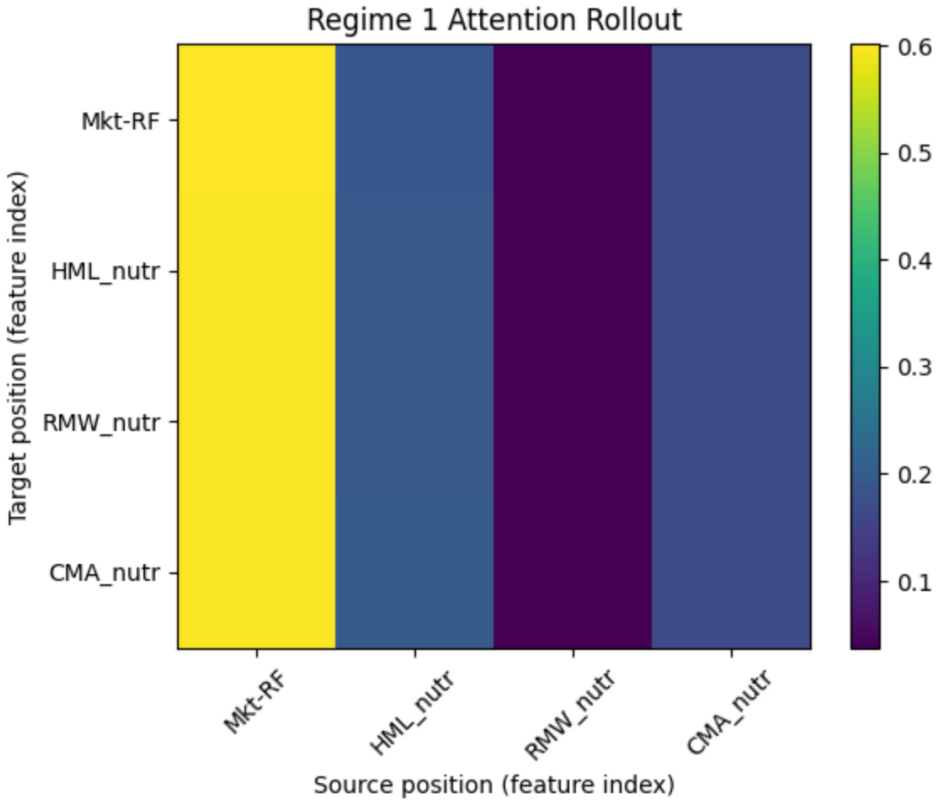
0.6
0.4
0.2


encoder_layer_1 | Head 3
Mkt-RF
HML_nutr
RMW_nutr
CMA_nutr
Target position (feature index)
Mkt-RF
HML_nutr
RMW_nutr
CMA_nutr
Source position (feature index)
0.30
0.28
0.26
0.24
0.22


Regime 0 Attention Rollout
Mkt-RF
HML_nutr
RMW_nutr
CMA_nutr
Target position (feature index)
Mkt-RF
HML_nutr
RMW_nutr
CMA_nutr
Source position (feature index)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15




```
Tested k=2: BIC=19531.68, Silhouette=0.2430
Tested k=3: BIC=19753.14, Silhouette=-0.0674
Tested k=4: BIC=19263.80, Silhouette=0.0247
Tested k=5: BIC=19558.47, Silhouette=-0.1136
Tested k=6: BIC=19459.73, Silhouette=-0.0465

Selected n_regimes = 2 via Silhouette (tie-break by BIC)

Regime counts:
Regime
0    1189
1     434
Name: count, dtype: int64
```